"Transcriptional Regulatory Networks Uncover SPI1 Regulatory Targets Important in Breast Cancer"

MCPHS UNIVERSITY

Bahareh Parcheh Bafi, George Acquaah-Mensah

Background

The Incidence rate of breast cancer among very young adults is becoming an object of public concern worldwide. There are a variety of presentations of breast cancer, with varying degrees of severity, and varying molecular presentations

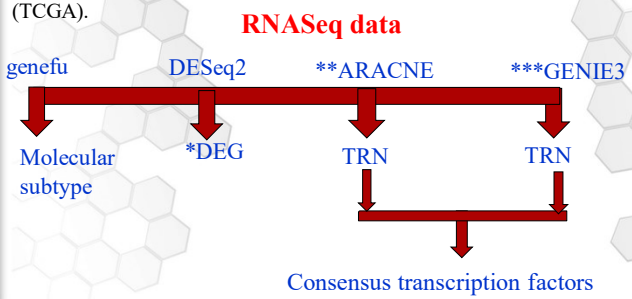
These studies were designed to gain some insights into the molecular pathways that play a major role in the transition from early- to advanced-stage cancer in young and mid-aged patients.

Methods

RNASeq data from:

- A South Korean study aged 35 years or younger
- A group of mostly white participants, 35 years-old or younger,
- A group of mid-aged patients, 50 to 60 years old

with either early- or late-stage tumors were leveraged from the International Cancer Genome Consortium or the cancer genome Atlas (TCGA).



*Using genes associated with GO

**Using genes involved in BRCA pathways (Hippo, NOTCH, PI3/Akt/NF-kB, Wnt/B-catenin signaling pathway, immune genes and oxidative stress)

***using all genes and selection top 2500

Total TCGA young	basal %	Her2 %	LumB %	LumA %	Normal %
35	20.0	11.4	51.4	11.4	5.7
Total TCGA 50to60	basal%	Her2%	LumB%	LumA%	Normal%
307	20	13	45	20	3
Total Korean	basal%	Her2%	LumB%	LumA%	Normal%
50	34.0	10	20	22	14

table. 1. MOLECULAR SUBTYPING

Results

SPI1 was showed up in ARACNE and GENIE3 derived networks consensus for all three groups. LAPTM5, C1QC as targets for SPI1 and CD4 for both SPI1 and FOXP3 are other important immune genes in these networks.

Previously, it was shown that SPI1 is important in Gastrointestinal cancer and can be a potential target for immunotherapy in this indication.

Molecular subtyping showed Korean patients mostly have basal-like molecular subtype whereas for mostly white TCGA, luminal subtypes are dominant.

To validate the targets, we used enrichment analysis using a consensus between ENCODE and ChEA databases, SPI1 over-represented among the targets.

SPI1 might be associated with cancer progression and high level of immune cells infiltration which can be potentially used for immunotherapy.

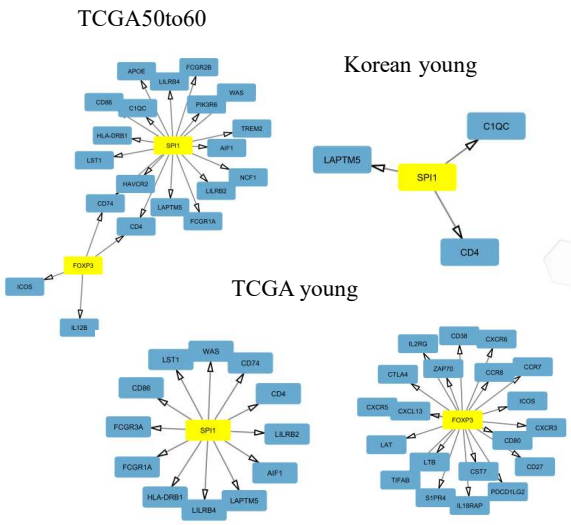


Fig 2. Cytoscape: ARACNE and GENIE3 consensus

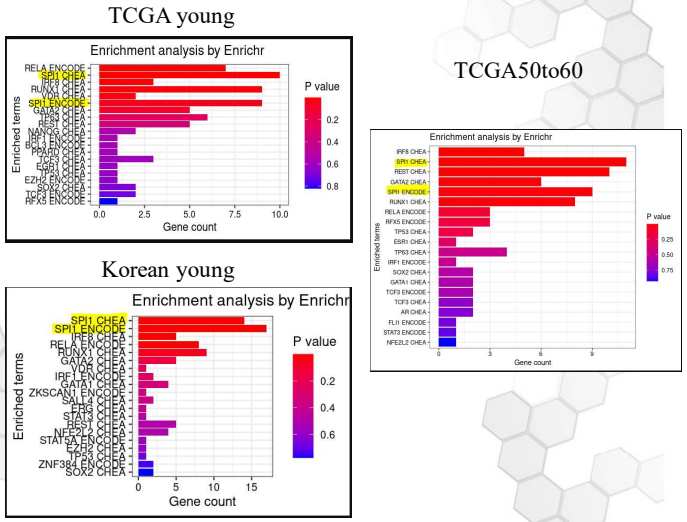


Fig. 2 . Enrichment plot: ENCODE and ChEA

Conclusion and discussion

in conclusion, comparing the gene expression profile in very young and mid-aged women with breast cancer in South Korea and the united states, with early-stage tumors and more advanced-stage tumors showed SPI1 might be an important gene in the transition from early- to late-stage in BrCA in younger and mid-aged patients. LAPTM5, C1QC and CD4 also can be interesting immune target genes involved in these regulatory networks

The Basal-like subtype is the leading type among BRCA Korean participants and luminal type is more dominant among the mostly white TCGA participants.

References

- Advanced Search | ICGC Data Portal
- Huang, J., Chen, W., Jie, Z., & Jiang, M. (2022). Comprehensive analysis of immune implications and prognostic value of spi1 in gastric cancer. *Frontiers in Oncology*, *12*. https://doi.org/10.3389/fonc.2022.820568
- Love MI, Huber W, Anders S (2014). "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2." *Genome Biology*, **15**, 550. doi: 10.1186/s13059-014-0550-8.

Disclosure: No one received any money for making this awesome poster.

MCPHS UNIVERSITY